The Law of the Wall

Problem

A measured data of a turbulent boundary layer is available in **data_turbulentboundarylayer.dat**. Based on measurements calculate y^+ and u^+ values and draw y^+ and u^+ values on the graph.

Also show how well the law of the wall correlates with the measurements and estimate y^+ values of different layers of the turbulent boundary layer. Measurements are made in a wind tunnel where η = 1.58·10^{-5} m^2/s and fluid is air.

Solution

Theory Background

Measurements show that, for both internal and external flows, the streamwise velocity in the flow near the wall varies logarithmically with distance from the surface. This behavior is known as the Jaw of the wall. A typical velocity profile for a turbulent profile is depicted on the Fig. 1 [1].

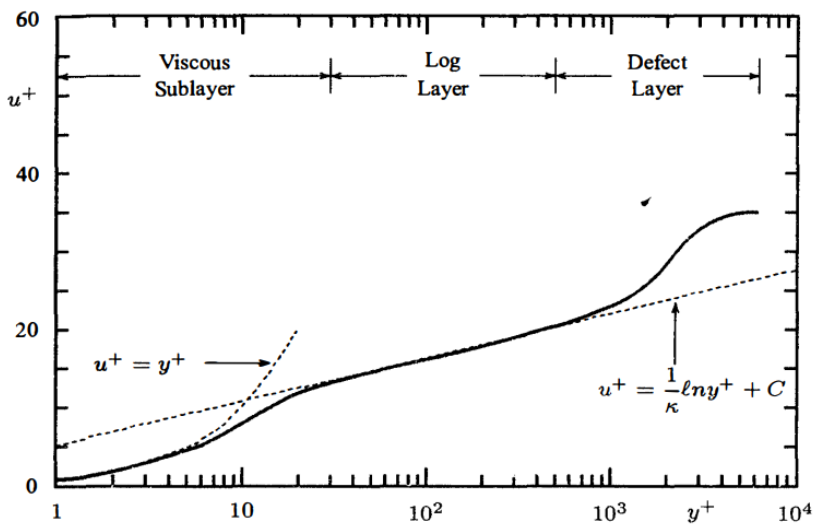

Figure 1: Typical velocity profile for a turbulent boundary layer

The graph displays the dimensionless velocity, u^+, and distance, y^+, defined as:

$$u^+ \equiv \frac{U}{u_\tau} \qquad \text{and} \qquad y^+ \equiv \frac{u_\tau y}{\nu}. \tag{1}$$

The quantity u_τ is known as the friction velocity, and is a velocity scale representative of velocities close to a solid boundary. U is the velocity of the fluid at a given distance y from the wall, while ν

represents dynamic viscosity [1].

Three layers can be distinguished: the viscous sublayer, the log layer and the defect layer. By definition, the log layer is the portion of the boundary layer where the sublayer and defect layer merge and the law of the wall accurately represents the velocity. It is not a distinct layer, but an overlap region between the inner and outer parts of the boundary layer [1].

Viscous sublayer: In viscous sublayer there is a linear relationship between velocity and wall distance, as follows:

$$u^+ = y^+. \tag{2}$$

Logarithmic-law layer: In this region u^+ is dependent on the logarithm of y^+, described by formula:

$$u^+ = \frac{1}{\kappa} \ln y^+ + C. \tag{3}$$

Correlation of measurements indicate C = 5.0 for smooth surfaces and $\kappa = 0.41$ for smooth and rough surfaces [1].

Analyis and Results

The data provided in the **data_turbulentboundarylayer.dat** represents two column y and U, respectively. For one to obtain the dimensionless quantities u^+ and y^+, u_τ has to be found first. To do that, one can start from viscous sublayer where u^+ is equal y^+. Combining equations from eq. (1), the expression for u_τ can be found as:

$$u_\tau = \sqrt{\frac{U\nu}{y}}. \tag{4}$$

Only first four points from given measurements shows the alignment in u_τ calculation with 0.2% of relative error comparing to the friction velocity of the first data point. Beyond that, the relative error rises above 20%, suggesting a possible transition between the viscous sublayer and the logarithmic layer. By averaging the friction velocity from these four points, the estimated u_τ is 1.2896 m/s. ν is 1.58 $\cdot 10^{-5}$ m^2/s.

Again, recalling eq. (1), $y^+_{measured}$ and $u^+_{measured}$ can be calculated by using $u_\tau = (1.2896 \pm 0.0026)$ m/s and $\nu = 1.58 \cdot 10^{-5}$ m^2/s.

$$y^+ = \frac{u_\tau y}{\nu}, \quad u^+_{measured} = \frac{U}{u_\tau}. \tag{5}$$

To incorporate predicted values for the logarithmic region, Eq. (3) has been applied

$$u^+_{predicted} = \frac{1}{\kappa} \ln y^+ + C. \tag{6}$$

For the first four points, y^+ has been directly assigned as the predicted values, as it was previously justified that these points lie within the viscous sublayer.

To differentiate between the logarithmic layer and the defect region, the relative error between $u^+_{measured}$ and $u^+_{predicted}$ can be analyzed. The relative error starts rising rapidly after reaching 2.5% discrepancy from the predicted value. That happens for $y^+ \approx 400$.

The final result is depicted on Fig. 2.

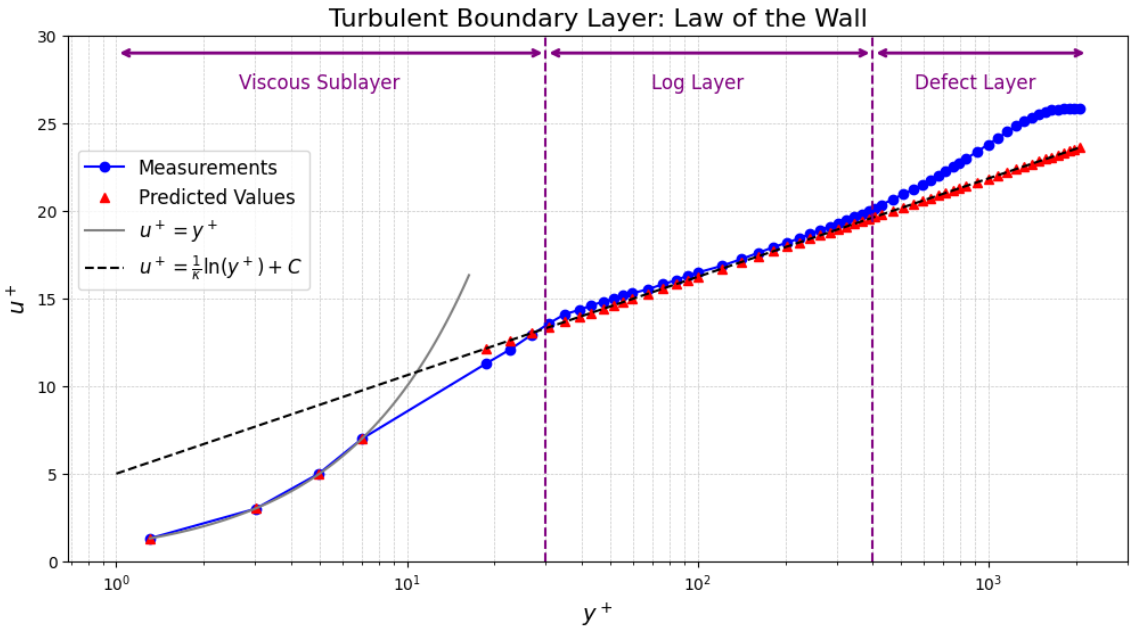

Figure 2: Velocity profile for turbulent boundary layer.

It can be observed that profile from the measurements aligns with theory predictions on Fig. 1. Data used for this report are presented in Tab..

Conclusion

The analysis of the turbulent boundary layer data confirms the validity of theoretical predictions within different flow regions. The friction velocity u_τ was estimated based on the viscous sublayer assumption, yielding a value of 1.2896 m/s. The comparison between $u^+_{measured}$ and $u^+_{predicted}$ indicates a strong agreement in the log layer, with a relative error below 2%. However, as y^+ increases beyond approximately 400, the relative error surpasses 2.5%, marking the transition to the defect layer.

The velocity profile in Fig. 2 shows that the measurements align well with the theoretical law of the wall, reinforcing the logarithmic relationship in the turbulent region. The findings highlight the effectiveness of classical turbulence models in predicting boundary layer behavior, while also emphasizing the importance of experimental validation.

y [m]	**U [m/s]**	u_τ **[m/s]**	$u_{\tau,relativeerror}$ **[%]**	y^+	$u^+_{measured}$	$u^+_{predicted}$	$u^+_{relativeerror}$
0,000016	1,6878	1,291	0	1,3059	1,3088	1,3059	0,2216
0,000037	3,873	1,286	0,39	3,0199	3,0033	3,0199	0,5527
0,000061	6,455	1,293	0,15	4,9788	5,0054	4,9788	0,5314
0,000086	9,037	1,2885	0,19	7,0193	7,0076	7,0193	0,167
0,000229	14,5475	1,0019	22,39	18,691	11,2806	12,1416	7,6326
0,000277	15,5871	0,9429	26,96	22,6088	12,0868	12,6057	4,2931
0,000328	16,6533	0,8957	30,62	26,7714	12,9135	13,0179	0,8085
0,000377	17,5063	0,8566	33,65	30,7708	13,575	13,3575	1,6022
0,000428	18,1894	0,8194	36,53	34,9335	14,1047	13,6669	3,1039
0,000479	18,4893	0,7809	39,51	39,0961	14,3372	13,9415	2,76
0,000527	18,8325	0,7514	41,8	43,0139	14,6034	14,1744	2,9377
0,000578	19,0957	0,7225	44,04	47,1765	14,8075	14,3997	2,754
0,000627	19,3023	0,6974	45,98	51,1759	14,9677	14,5982	2,4686
0,000678	19,5522	0,675	47,71	55,3385	15,1614	14,7889	2,4569
0,000728	19,7088	0,654	49,34	59,4195	15,2829	14,9625	2,0965
0,000828	20,0287	0,6182	52,11	67,5816	15,5309	15,2764	1,6387
0,000928	20,3818	0,5891	54,37	75,7436	15,8047	15,5545	1,5831
0,00103	20,6751	0,5632	56,37	84,0689	16,0322	15,8089	1,3928
0,00113	21,0216	0,5422	58	92,2309	16,3009	16,0349	1,6318
0,00123	21,2548	0,5225	59,53	100,3929	16,4817	16,2417	1,4562
0,00148	21,7346	0,4817	62,69	120,798	16,8538	16,693	0,9541
0,00173	22,2378	0,4507	65,09	141,203	17,244	17,0737	0,9876
0,00198	22,7009	0,4256	67,03	161,6081	17,6031	17,4029	1,1373
0,00223	23,1141	0,4047	68,65	182,0132	17,9235	17,6929	1,2866
0,00248	23,444	0,3865	70,06	202,4182	18,1793	17,952	1,2503
0,00273	23,7472	0,3707	71,29	222,8233	18,4144	18,1863	1,2387
0,00298	24,087	0,3574	72,32	243,2284	18,6779	18,4	1,4879
0,00323	24,3769	0,3453	73,25	263,6334	18,9027	18,5965	1,6199
0,00348	24,6168	0,3343	74,11	284,0385	19,0887	18,7783	1,6261
0,00373	24,87	0,3246	74,86	304,4435	19,285	18,9475	1,7501
0,00398	25,1033	0,3157	75,55	324,8486	19,466	19,1058	1,8504
0,00423	25,3565	0,3078	76,16	345,2537	19,6623	19,2543	2,075
0,00448	25,5631	0,3003	76,74	365,6587	19,8225	19,3944	2,1597
0,00473	25,7997	0,2936	77,26	386,0638	20,006	19,5268	2,3953
0,00498	25,9863	0,2871	77,76	406,4689	20,1507	19,6525	2,4724
0,00523	26,2262	0,2815	78,2	426,8739	20,3367	19,7719	2,7772
0,00573	26,6227	0,2709	79,02	467,6841	20,6442	19,9946	3,1466
0,00623	27,0159	0,2618	79,72	508,4942	20,9491	20,1987	3,582
0,00673	27,3391	0,2533	80,38	549,3043	21,1997	20,387	3,8335
0,00723	27,7122	0,2461	80,94	590,1144	21,489	20,5617	4,3152
0,00773	28,0255	0,2393	81,46	630,9246	21,7319	20,7248	4,6342
0,00823	28,3986	0,2335	81,91	671,7347	22,0212	20,8777	5,1927
0,00873	28,6885	0,2279	82,35	712,5448	22,246	21,0216	5,5039
0,00923	29,0084	0,2228	82,74	753,3549	22,4941	21,1574	5,9424
0,00973	29,3049	0,2181	83,11	794,1651	22,724	21,2861	6,3277
0,0102	29,5948	0,2141	83,42	832,5266	22,9488	21,4011	6,7441
0,0112	30,1479	0,2062	84,03	914,1468	23,3777	21,6292	7,4793
0,0122	30,6644	0,1993	84,56	995,7671	23,7782	21,8378	8,1604
0,0132	31,1509	0,1931	85,04	1077,387	24,1555	22,03	8,7992
0,0142	31,6107	0,1875	85,48	1159,008	24,512	22,2081	9,3991
0,0152	32,0205	0,1824	85,87	1240,628	24,8298	22,3741	9,8901
0,0162	32,3704	0,1777	86,24	1322,248	25,1011	22,5295	10,245
0,0172	32,6836	0,1733	86,58	1403,868	25,344	22,6756	10,5287
0,0182	32,9235	0,1691	86,9	1485,489	25,53	22,8134	10,6408
0,0192	33,0934	0,165	87,22	1567,109	25,6618	22,9439	10,5912
0,0202	33,2034	0,1612	87,51	1648,729	25,7471	23,0677	10,4066
0,0212	33,2567	0,1574	87,81	1730,349	25,7884	23,1856	10,0929
0,0222	33,29	0,1539	88,08	1811,97	25,8142	23,298	9,7473
0,0232	33,3033	0,1506	88,33	1893,59	25,8245	23,4054	9,3675
0,0242	33,32	0,1475	88,57	1975,21	25,8375	23,5084	9,0144
0,0252	33,3133	0,1445	88,81	2056,83	25,8323	23,6071	8,614

Table 1: Data used for the calculations.

References

[1] DC Wilcox. Turbulence modeling for cfd. *DCW industries, La Canada*, 1998.